Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Certified Lean Six Sigma Black Belt (CLSSBB) Training

TGS Ref No: TGS-2024051900

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 3

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1	1 June 2026	Initial release - CLSSBB lesson plan.	Dr. Alfred Ang
2	20 July 2026	Major revision: rebuilt as a 5-day Black Belt progression from the Green Belt course (CLSSGB). DMAIC is recapped IN DEPTH (a Black Belt must be able to mentor Green Belts through these tools) and then extended with Black-Belt-only content the Green Belt course does not teach: attribute MSA and Kappa analysis, nested and destructive gauge studies, non-normal capability and Cpk versus Ppk, multiple regression with model diagnostics and VIF, ANOVA with interaction effects, non-parametric tests, multi-vari studies, full and fractional factorial Design of Experiments with confounding and resolution, response surface methodology, Taguchi robust design, CUSUM, EWMA, short-run and multivariate SPC, change leadership and resistance management, financial benefits validation, and mentoring Green Belts. Expanded to 34 hands-on labs across five days. THE CAPSTONE PROJECT RUNS FROM DAY 1 - each learner selects a real workplace process in Lab 1 and every subsequent lab adds one artifact to it; Day 5 consolidates the artifacts into an A3	Dr. Alfred Ang

		storyboard and a steering committee presentation.	
3	20 July 2026	Corrected the assessment model to match the registered WSQ instrument: the assessment is the Written Assessment (WA) Short-Answer Questions plus the Case Study (CS), both open book. The capstone project and steering committee presentation are retained as an ADDITIONAL course deliverable on top of the WSQ assessment, not as a replacement for it. Added the online SIPOC tool (alfredang.github.io/sipoc) to the toolkit, Lab 6 and the Define slides.	Dr. Alfred Ang

TABLE OF CONTENTS

Course Information

Course Title	Certified Lean Six Sigma Black Belt (CLSSBB) Training
WSQ Course Reference	TGS-2024051900
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	5 days · 8 training hours per day (40 hours)
Daily Timing	9:30 am – 6:30 pm (1-hour lunch; tea breaks within training time)
Mode	Instructor-led, hands-on Lean Six Sigma labs applied to each learner's OWN workplace capstone project (Meridian Medical Devices scenario provided as a fallback)
TSC Alignment	Quality Process Control (ELE-QUA-5006-1.1)
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Lead a Lean Six Sigma project portfolio — select, charter and govern projects aligned to business strategy (A1, A2).
- LO2: Validate measurement systems for continuous AND attribute data using Gage R&R and Kappa analysis (A4).
- LO3: Analyse process data using multiple regression, ANOVA, non-parametric tests and multi-vari studies (A3).
- LO4: Design, run and interpret experiments using full and fractional factorial DOE and response surface methods (A3, K2).
- LO5: Select, de-risk and pilot improvements, and validate financial benefits with Finance sign-off (A5).
- LO6: Sustain the gain using advanced SPC — CUSUM, EWMA, short-run and multivariate control charts (A5, K1).
- LO7: Lead change and mentor Green Belts through resistance, stakeholder management and tollgate governance (A1, K1).
- LO8: Deliver a capstone improvement project and present it to a steering committee (A1-A5, K1, K2).

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 2 questions, 1 hour, open book (K1, K2).
- Case Study (CS) — an applied DMAIC case study, 90 minutes, open book (A1-A5).
- Capstone Project & Presentation — an ADDITIONAL course deliverable on top of the WSQ assessment: the consolidated DMAIC storyboard built across Labs 1-30, presented and defended to a steering committee on Day 5.
- Format: Open Book — course slides, Learner Guide and approved materials only.

- The WSQ assessment is conducted on Day 5: WA (SAQ) from 4:00 pm (1 hour), then the Case Study from 5:00 pm (90 minutes).
- The capstone presentation is delivered on Day 5 and is additional to the WSQ assessment.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

Course Schedule

Day 1 — The Black Belt role, portfolio selection, capstone chartering and Define

Time	Duration	Topic / Activity
9:30–10:00	30 min	Welcome, course introduction, ground rules and mandatory digital attendance (AM). Briefing on the capstone project: every learner selects a real workplace process today and builds it across all five days [slides 2–22]
10:00–11:15	75 min	**FOUNDATIONS — the Black Belt role versus the Green Belt role; leading a portfolio; owning the advanced statistics; mentoring Green Belts; project portfolio selection criteria; strategic (Hoshin) alignment; Cost of Poor Quality at portfolio scale [slides 23–93]**
11:15–11:30	15 min	Tea break
11:30–12:30	60 min	**FOUNDATIONS — DMAIC recap at depth: the purpose, core tools, tollgate deliverable and common failure mode of each phase; Y = f(X); sigma level, DPMO and the 1.5 sigma shift; tollgate governance**
12:30–13:30	60 min	Lunch break
13:30–15:15	105 min	Hands-on: Lab 1: Select and Charter Your Capstone Project; Lab 2: Portfolio Selection, Strategy Alignment and the Black Belt Role; Lab 3: DMAIC at Depth — Recap, Y = f(X) and Baseline Sigma for Your Project [slides 94–105]
15:15–15:30	15 min	Tea break
15:30–17:00	90 min	**DMAIC · DEFINE — VOC, affinity and Kano in depth; CTQ trees, operational definitions and specification limits; CTQ FLOWDOWN from business Y to controllable Xs; charter**

		governance; stakeholder influence/support mapping; resistance diagnosis; ADKAR and Kotter change leadership [slides 106–179]
17:00–18:15	75 min	Hands-on: Lab 4: Voice of the Customer, Affinity and Kano — In Depth; Lab 5: CTQ Trees and CTQ Flowdown to Controllable Xs [slides 180–187]
18:15–18:30	15 min	Day 1 recap, Q&A and PM digital attendance

Total training time: 480 minutes (8 hours).

Day 2 — Measure — advanced measurement systems, attribute MSA and capability

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 1 recap and mandatory digital attendance (AM)
9:45–11:00	75 min	Hands-on: Lab 6: SIPOC, Process Mapping and Scope Governance; Lab 7: Stakeholder Analysis, Resistance Management and Change Leadership [slides 188–196]
11:00–11:15	15 min	Tea break
11:15–12:30	75 min	**DMAIC · MEASURE — process mapping, swimlanes and handoff analysis; value stream mapping with material and information flow; the timeline ladder and value-added ratio; takt time; the eight wastes (DOWNTIME); data types and operational definitions; the data collection plan [slides 197–290]**
12:30–13:30	60 min	Lunch break
13:30–15:00	90 min	Hands-on: Lab 8: Value Stream Mapping, Takt Time and the Eight Wastes — In Depth; Lab 9: Data Types, Operational Definitions and the Data Collection Plan [slides 291–298]
15:00–15:15	15 min	Tea break
15:15–16:45	90 min	**DMAIC · MEASURE — sampling schemes, sample size formulas and RATIONAL SUBGROUPING; continuous Gage R&R in depth (%study variation, ndc); ATTRIBUTE MSA and KAPPA analysis; nested and destructive**

		gauge studies; yield, RTY and DPMO; non-normal capability, Box-Cox, Cpk versus Ppk
16:45–18:15	90 min	Hands-on: Lab 10: Sampling Strategy, Sample Size and Rational Subgrouping; Lab 11: Continuous Gage R&R — In Depth; Lab 12: Attribute MSA and Kappa Analysis (Black Belt Only) [slides 299–310]
18:15–18:30	15 min	Day 2 recap and PM digital attendance

Total training time: 480 minutes (8 hours).

Day 3 — Analyze — multiple regression, ANOVA, non-parametric and multi-vari

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 2 recap and mandatory digital attendance (AM)
9:45–11:00	75 min	Hands-on: Lab 13: Nested and Destructive Gauge Studies (Black Belt Only); Lab 14: Baseline Capability — Cp, Cpk, Pp, Ppk and Non-Normal Data [slides 311–319]
11:00–11:15	15 min	Tea break
11:15–12:30	75 min	**DMAIC · ANALYZE — common versus special cause and tampering (the Deming funnel); run charts and the six non-random patterns; Pareto weighted by cost; stratification and boxplots; fishbone 5M+E; 5 Whys; multi-voting and the cause-and-effect matrix [slides 320–419]**
12:30–13:30	60 min	Lunch break
13:30–14:45	75 min	Hands-on: Lab 15: Variation, Run Charts, Pareto and Stratification — In Depth; Lab 16: Fishbone, 5 Whys and Cause Prioritisation — In Depth [slides 420–427]
14:45–15:00	15 min	Tea break
15:00–16:45	105 min	**DMAIC · ANALYZE — hypothesis testing in depth (alpha, beta, POWER, test selection logic, assumption checking); MULTIPLE REGRESSION and model diagnostics (adjusted R-squared, residual plots, multicollinearity and VIF); ANOVA and alpha inflation; two-way ANOVA and INTERACTION effects; non-parametric tests; MULTI-**

		VARI studies; chi-square
16:45–18:15	90 min	Hands-on: Lab 17: Hypothesis Testing — In Depth, With Test Selection; Lab 18: Multiple Regression and Model Diagnostics (Black Belt Only); Lab 19: ANOVA, Interactions and Non-Parametric Tests (Black Belt Only) [slides 428–441]
18:15–18:30	15 min	Day 3 recap and PM digital attendance

Total training time: 480 minutes (8 hours).

Day 4 — Improve and Control — DOE, RSM and advanced SPC

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 3 recap and mandatory digital attendance (AM)
9:45–11:00	75 min	Hands-on: Lab 20: Multi-Vari Studies and Chi-Square Analysis (Black Belt Only); Lab 21: Analyze Tollgate — Proving Root Cause and Telling the Data Story [slides 442–451]
11:00–11:15	15 min	Tea break
11:15–12:30	75 min	**DMAIC · IMPROVE — solution generation and benchmarking; Lean countermeasures (5S, poka-yoke, pull, JIT, Heijunka, Jidoka, standard work); the weighted solution selection matrix; FMEA, RPN, the severity override and re-scoring; DESIGN OF EXPERIMENTS: why OFAT fails, full factorial 2^k, randomisation, replication, blocking, main effects and interaction plots [slides 452–540]**
12:30–13:30	60 min	Lunch break
13:30–15:00	90 min	Hands-on: Lab 22: Solution Generation, Selection and Lean Countermeasures — In Depth; Lab 23: Full Factorial Design of Experiments (Black Belt Only) [slides 541–549]
15:00–15:15	15 min	Tea break
15:15–16:45	90 min	**DMAIC · IMPROVE and CONTROL — fractional factorial designs, confounding, alias structure and RESOLUTION III/IV/V; fold-over designs; response surface methodology, centre points, central composite designs; Taguchi robust**

		design; SPC recap and chart selection; CUSUM, EWMA, short-run and MULTIVARIATE SPC; control plan governance; financial benefits validation [slides 566–639]
16:45–18:15	90 min	Hands-on: Lab 24: Fractional Factorial Designs, Confounding and Resolution (Black Belt Only); Lab 25: Response Surface Methodology and Robust Design (Black Belt Only); Lab 26: FMEA, Pilot Design and Implementation Planning — In Depth; Lab 27: Statistical Process Control and Chart Selection — In Depth [slides 550–643]
18:15–18:30	15 min	Day 4 recap and PM digital attendance

Total training time: 480 minutes (8 hours).

Day 5 — Capstone project consolidation and steering committee presentation

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 4 recap and mandatory digital attendance (AM)
9:45–11:00	75 min	Hands-on: Lab 28: CUSUM, EWMA and Short-Run SPC (Black Belt Only); Lab 29: Multivariate SPC and Control Plan Governance (Black Belt Only) [slides 644–653]
11:00–11:15	15 min	Tea break
11:15–11:45	30 min	Hands-on: Lab 30: Financial Benefits Validation, Handover and Project Closure [slides 654–659]
11:45–12:30	45 min	**CAPSTONE — consolidating your artifacts from Labs 1-30; the A3 storyboard structure (Background, Current State, Goal, Root Cause, Countermeasures, Results, Follow-up); building the steering committee presentation; the two-minute rule and the appendix strategy [slides 660–697]**
12:30–13:30	60 min	Lunch break
13:30–14:15	45 min	Hands-on: Lab 31: Consolidating Your Capstone — The A3 Storyboard; Lab 32: Building the Steering Committee Presentation [slides 698–705]
14:15–15:15	60 min	Hands-on: Lab 33: Capstone Presentation and Defence

		[slides 706–709] — capstone presentations to the steering committee panel
15:15–15:30	15 min	Tea break
15:30–15:45	15 min	Hands-on: Lab 34: Lessons Learned, Mentoring Green Belts and Your Black Belt Journey [slides 710–714] — lessons learned, mentoring Green Belts and your 90-day plan
15:45–16:00	15 min	**Course recap and Briefing for Assessment [slides 715–726]**
16:00–17:00	60 min	**WSQ ASSESSMENT — Written Assessment (WA), Short-Answer Questions. 1 hour, open book**
17:00–18:30	90 min	**WSQ ASSESSMENT — Case Study (CS). 90 minutes, open book. PM digital attendance**

Total training time: 480 minutes (8 hours).

Lab Reference (aligned to the DMAIC phases)

DMAIC phase / Topic	Weighting	Labs
FOUNDATIONS: The Black Belt Role, Portfolio Selection & Your Capstone	10%	Lab 1, Lab 2, Lab 3
DEFINE: Define at Black Belt Depth — Charter, Stakeholders & Change	12%	Lab 4, Lab 5, Lab 6, Lab 7
MEASURE: Measure at Black Belt Depth — Advanced Measurement Systems	16%	Lab 8, Lab 9, Lab 10, Lab 11, Lab 12, Lab 13, Lab 14
ANALYZE: Analyze at Black Belt Depth — Advanced Statistical Analysis	22%	Lab 15, Lab 16, Lab 17, Lab 18, Lab 19, Lab 20, Lab 21
IMPROVE: Improve at Black Belt Depth — Design of Experiments	18%	Lab 22, Lab 23, Lab 24, Lab 25, Lab 26
CONTROL: Control at Black Belt Depth — Advanced SPC & Sustaining	12%	Lab 27, Lab 28, Lab 29, Lab 30
CAPSTONE: Capstone Project & Steering Committee Presentation	10%	Lab 31, Lab 32, Lab 33, Lab 34